# Tests & Diagnostics — AGEL0206 $\sigma$_e ApJL paper

Last updated: 2026-04-30 Headline: $\sigma$_e(<R_e) = 268 ± 32 km/s (stat ±24 ⊕ I-shape ±13 ⊕ mask ±16 ⊕ frame ±5 ⊕ centering ±4) Method: §6cum cumulative I-weighted ppxf at R<R_e=2.305" inside the F200LP-masked, frame-aware, SPS-pooled (FSPS+EMILES+XSL) bootstrap × polynomial-degree posterior

This document is the canonical index of every test, sweep, audit, and sensitivity check run for the $\sigma$_e measurement. Each row points to the script/notebook that runs it and the result file or note it produced.

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## 0. Quick-reference test matrix

| # | Test | Where | Result | Status |
|---|---|---|---|---|
| A. Foundational | | | | |
| A1 | KCWI cube provenance | `NOTES_methodology_2026-04-27.md` | Multi-night Aug/Sept/Dec 2025; header mislabel only | ✓ |
| A2 | Systemic redshift via line fitting | `notebooks/04_redshift_verification.ipynb`, `scripts/redshift_verify.py` | z = 0.67564 (vs DR2 0.67511) | ✓ |
| A3 | R_e from masked curve-of-growth (F140W+F200LP) | `scripts/measure_Re.py`, `scripts/final_sigma_e.py:curve_of_growth` | R_e = 2.305" = 16.23 kpc | ✓ |
| A4 | HST-mean center via centroid_2dg | `scripts/final_sigma_e.py:find_center` | F140W & F200LP agree to 0.36" | ✓ |
| A5 | Bagpipes SED fitting (M⋆) | `notebooks/02_streamlined_Bagpipes_SED.ipynb`, `08_sersic_total_photometry.ipynb` | log M⋆ = 11.33 +0.07/—0.09 | ✓ |
| B. Pipeline correctness audits | | | | |
| B1 | Instrumental LSF audit (DISPSCAL=0.294) | `scripts/ifu_spectral_resolution.py` | FWHM=0.692 Å; $\sigma$_v_inst≈12-14 km/s | ✓ |
| B2 | $\sigma$_inst sensitivity (×0.5 to ×2.0 LSF) | `scripts/sigma_inst_sensitivity.py`, audit 3 | max \|Δ$\sigma$\| = 0.83 km/s | ✓ |
| B3 | SPS template wavelength frames | `scripts/audit_ppxf_methodology.py`, NOTES Test 2 | FSPS=vacuum, EMILES=air, XSL=air (Ca K minimum + V_sys) | ✓ |
| B4 | V_sys air vs vacuum × 5 polynomial degrees | `scripts/audit_ppxf_methodology.py` audit 1 | ΔV consistent within ±2 km/s across degrees | ✓ |
| B5 | z × air-vac differential (obs vs rest application) | `scripts/audit_ppxf_methodology.py` audit 2 | 1.83 km/s differential — sub-budget | ✓ |
| B6 | fwhm_gal_dict frame consistency check | `scripts/audit_ppxf_methodology.py` audit 4 | dict in REST frame, matches Cappellari pattern | ✓ |
| C. SPS templates & frame-fix | | | | |
| C1 | Frame-aware ppxf: per-SPS native frame | `scripts/bootstrap_ppxf.py:SPS_NATIVE_FRAME` + `frame_galaxy='auto'` | V_sys split collapsed from ~110 to ~15 km/s | ✓ |
| C2 | End-to-end V_sys closure (frame swap) | NOTES Test 3, addendum 2026-04-29 | Frame fix matches diagnosis | ✓ |
| C3 | $\sigma$ shift from frame fix | NOTES addendum, `error_budget()` | ≤5 km/s across SPS → carried as ±5 km/s budget | ✓ |
| C4 | 3-SPS pooled posterior | `scripts/final_sigma_e.py:pool_sps` | Per-SPS V_sys subtracted before pooling | ✓ |
| D. Aperture & I(r) | | | | |
| D1 | §6cum cumulative I-weighted ppxf | `scripts/final_sigma_e.py:run_aperture_sps` | $\sigma$_e(<R_e) = 267.82 km/s headline | ✓ |

| # | Test | Where | Result | Status |
|---|---|---|---|---|
| D2 | §6cum vs §7 (annular) cross-check | `notebooks/07c`, `~/.claude/.../reference_cumulative_vs_annular_sigma_e.md` | §7=255, §7b=271 — all <1$\sigma$ from §6cum | ✓ |
| D3 | I(r) shape sweep (10 shapes, fixed mask) | `scripts/run_isource_shape_sweep.py`, `scripts/analyze_isource_shape_sweep.py` | Range = 12.9 km/s (excluding F200LP_Sersic2D outlier) → $\pm$13 budget | ✓ |
| D4 | I(r) shape sweep refresh post-frame-fix at N=500 | NOTES addendum (d), 2026-04-29 | Frame fix has <0.5 km/s impact; $\pm$13 budget validated | ✓ |
| D5 | Aperture choice: R<R_e/8, R<R_e/2, R<R_e | `scripts/final_sigma_e.py:APERTURE_FRACS` | R<R_e/8 dropped (inside seeing FWHM/2 = 0.64”) | ✓ |
| D6 | Equal-N vs equal-width annular binning | `notebooks/07b` (5-bin), `notebooks/07c` (equal-N) | Equal-N inside R_safe=3R_e/4 chosen | ✓ |
| D7 | R_e source systematic (mean vs F140W vs F200LP vs CaHK+G) | `scripts/final_sigma_e.py` §7, `notebooks/09 §7b` | Spread = 16.9 km/s (mean=268, F140W=265, F200LP=276, CaHK=282); sub-budget | ✓ |
| E. Mask treatment | | | | |
| E1 | F200LP arc mask reprojected to IFU grid | `scripts/final_sigma_e.py:load_setup` | 0.7% of all spaxels flagged (~38 inside R<R_e) | ✓ |
| E2 | Hard mask (w=0.0) headline | `scripts/final_sigma_e.py`, track A | $\sigma$_e = 267.82 $\pm$ 24 km/s | ✓ |
| E3 | No-mask sensitivity (w=1.0) | track B | $\sigma$_e = 252.82, $\Delta = -15.0$ km/s | ✓ |
| E4 | Soft-mask single-point (w=0.5) | `scripts/soft_mask_track.py`, track C, NOTES addendum (b) | $\sigma$_e = 258.54, $\Delta = -9.29$; super-linear | ✓ |
| E5 | 5-point mask-weight sweep w∈{0,0.25,0.5,0.75,1} | `scripts/soft_mask_track.py --weight ...`, `scripts/analyze_mask_weight_sweep.py`, NOTES addendum (c) | Per-step drops 5.0→4.3→3.5→2.2; quadratic c=+7.3 (concave-up); threshold-dominated | ✓ |
| E6 | Mask dilation (over-masking diagnostic) | `notebooks/07d_sigma_e_forceful_mask.ipynb`, `~/.claude/.../project_nb07d_overmasking_finding.md` | Dilation inflates $\sigma$ via noise; do NOT dilate (negative finding) | ✓ |
| E7 | Arc-spectrum subtraction sibling | `notebooks/07e_sigma_e_arc_subtract.ipynb` | Matches §6cum within 0.1 km/s — residual arc dilution sub-dominant | ✓ |
| E8 | Arc-as-sky mechanism test (no-mask + arc-sky template) | `scripts/run_07f_arc_sky.py`, `notebooks/07f_arc_sky_subtract.ipynb` | $\sigma$_D=274.6 vs $\sigma$_A=267.8 / $\sigma$_B=252.8; recovery=145% — dilution explains the full no-mask $\Delta$ | ✓ |
| F. Centering | | | | |
| F1 | 5-center sweep ($\pm$0.4” perturbations) | `NOTES_centering_investigation_2026-04-27.md`, `scripts/regen_s6cum_nomask_diagnostic.py` | $\sigma$_e spread = 3.7 km/s → $\pm$4 km/s budget | ✓ |
| F2 | HST F140W vs F200LP centroid offset | `scripts/final_sigma_e.py:find_center` | 0.36” — driven by F200LP arc; HST-mean robust | ✓ |
| G. Bootstrap & polynomial degree | | | | |
| G1 | Wild bootstrap (Rademacher $\times$ local residual) | `scripts/bootstrap_ppxf.py:run_bootstrap_single_degree` | 75-pix rolling window for $\sigma$_loc | ✓ |

| # | Test | Where | Result | Status |
|---|---|---|---|---|
| G2 | Parallel bootstrap (joblib, BLAS=1 per worker) | `scripts/bootstrap_ppxf_parallel.py` | ~6 min for 6 fits at N=500 | ✓ |
| G3 | Polynomial degree sweep (deg 15-29) | `notebooks/03c`, `scripts/build_nb09.py:§6.5` | $\sigma$ flat within bootstrap envelope; polynomial saturated | ✓ |
| G4 | N=50 smoke vs N=500 production agreement | `scripts/final_sigma_e.py --n_bootstrap` | Within 1 km/s | ✓ |
| H | Cross-checks (independent estimators) | | | |
| H1 | §7 discrete Gültekin annular sum | `notebooks/07c §7` | 254.99 −24.2/+28.4 km/s (arc-filtered) | ✓ |
| H2 | §7b flat-$\sigma$ extrapolation into outer annulus | `notebooks/07c §7b` | 271 −33/+35 km/s | ✓ |
| H3 | F200 mask sensitivity at N=500 | `results/annular_bootstrap_07c_nomask/` | Δ_mask = −16.4 km/s → ±16 budget | ✓ |
| I. Earlier sigma-discrepancy work (Tests 1-21) | | | | |
| I1 | NB01 vs NB05 $\sigma$ reproduction | `notebooks/05x` Tests 1-4 | Confirmed $\sigma$ depends on aperture | ✓ |
| I2 | Source mask + PSF-aware masking | `notebooks/05x` Tests 7-10 | F140W mask flags deflector core; F200LP mask is the right arc mask | ✓ |
| I3 | Threshold sweep + contamination weighting | `notebooks/05x` Tests 11-13 | Pre-Gültekin masking strategy work | ✓ |
| I4 | V_rms profile + bootstrap | `notebooks/05x` Tests 14-16 | Pre-Gültekin radial work | ✓ |
| I5 | Redshift sensitivity (z=0.67511 vs 0.67564) | `notebooks/05x` Test 17 | <1 km/s effect on $\sigma$ | ✓ |
| I6 | Voronoi binning attempt | `notebooks/05x` Test 18 | Failed — abandoned | ✓ |
| I7 | PowerBin spatial binning | `notebooks/05x` Test 19, `scripts/run_powerbin_test19.py` | Rejected as binning scheme — outer-bin $\sigma$>800 km/s | ✓ |
| I8 | R_e four-way comparison | `notebooks/05x` Test 20, `scripts/measure_Re.py` | F140W+F200LP masked CoG mean = 2.305" headline | ✓ |
| I9 | Per-spaxel vs PowerBin rotation map | `notebooks/05x` Test 21 | No rotation on Re-scale; $\sigma$-dominated | ✓ |

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## 1. Headline numbers

$\sigma$_e at three apertures (cumulative I-weighted ppxf, frame-aware, masked)

| Aperture | $\sigma$_e [km/s] | Notes |
|---|---|---|
| R<R_e (=2.305") | 268 ± 32 (stat ±24) | paper headline — KH13/Greene+20 $\sigma$_e |
| R<R_e/2 (=1.152") | 224 ± 18 | gradient diagnostic |
| R<R_e/8 (=0.288") | dropped | inside seeing FWHM/2 = 0.64" |

Error-budget composition (R<R_e)

| Component | Value | Source |
|---|---|---|
| Statistical (bootstrap pooled 1$\sigma$) | ±24 | nb09, this paper |
| I-shape spread (10 shapes excluding outlier) | ±13 | nb07c sweep, refreshed N=500 post-frame-fix |
| Mask on/off Δ | ±16 | nb07c, nb09 §5 |
| Frame fix (max $\sigma$ shift across SPS) | ±5 | NOTES addendum + audit 1 |

| Component | Value | Source |
|---|---|---|
| Centering (5-center sweep, $\pm$0.4”) | $\pm$4 | `NOTES_centering_investigation_2026-04-27.md` |
| Quadrature total | $\pm$32 | `scripts/final_sigma_e.py:error_budget` |

Sensitivity values (for paper text)

- Soft-mask (w=0.5) at R<R_e: 258.5 km/s — $\Delta$ from headline = −9.3 km/s
- No-mask (w=1.0) at R<R_e: 252.8 km/s — $\Delta$ from headline = −15.0 km/s
- $\sigma$_e(w) is essentially linear with weak super-linearity (quadratic c=+7.3)

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## 2. Detailed test catalog

This expands each row of the matrix above. Subsections reference scripts, notebooks, and result files — open the linked file for the detailed code and the inline plots/output.

### A. Foundational data and geometry

A1 KCWI cube provenance — The cube `Nov17_2025_DESJ0206_RL_combined_icubes_wcs.fits` is the final multi-night reduction (Aug 29 K409 + Sept 29 K409 + Dec 29 + Nov 17 2025). The header is mislabeled (DATE-OBS/PROGNAME/PROGPI describe only the last stacking pass). Settled in `NOTES_methodology_2026-04-27.md`. Cite the multi-night provenance, not the header.

A2 Systemic redshift — Independent line-by-line Gaussian fitting of Ca H+K, [OII], G-band, etc. gives z_systemic = 0.67564 (NIST air wavelengths). DR2 catalog z=0.67511 (used by older notebooks 03/03b) is offset by ~95 km/s from the line-fit value; nb09 uses 0.67564 throughout. Notebook: `notebooks/04_redshift_verification.ipynb` Module: `scripts/redshift_verify.py` (provides ABSORPTION_LINES / EMISSION_LINES dicts, `gaussian_fit`, `multi_line_fit`, `oii_doublet_fit`)

**A3 Effective radius R_e = 2.305“** — Mean of F140W and F200LP masked curve-of-growth values (2.168 and 2.441”). The masked CoG zeros mask-flagged HST pixels before the radial integral so the arc/companion contributions don’t bias the bright-end half-light point. Computed in `scripts/measure_Re.py` and re-derived in `scripts/final_sigma_e.py:curve_of_growth` for the nb09 pipeline. Cross-check: IFU white-light proper-masked CoG gave 2.61” earlier — the HST-based 2.305” is now the headline.

A4 HST-mean center — `photutils.centroid_2dg` on F140W and F200LP cutouts (with their own `_mask.fits` as the mask) gives sub-pixel centroids in each HST WCS. We average in world coordinates, then propagate to IFU sub-pixel: `scripts/final_sigma_e.py:find_center`. F140W vs F200LP centroid offset = 0.36” — flags the F200LP arc-affected detection but the HST-mean is robust (verified by the 5-center sweep, F1).

A5 Stellar mass — Bagpipes SED fitting on HST + JWST aperture photometry → log M⋆/M⊙ = 11.33 +0.07/−0.09 (notebook 02). Cross-checked at the Sersic-total level in `notebooks/08_sersic_total_photometry.ipynb`.

### B. Pipeline correctness audits (**`scripts/audit_ppxf_methodology.py`**)

Four orthogonal correctness audits run end-to-end on the headline aperture spectrum to verify ppxf usage matches Cappellari (2017, 2023). Saved in `results/ppxf_methodology_audit.npz`. Documented in `NOTES_nb09_frame_fix_and_final_sigma_e_2026-04-28.md` ADDENDUM 2026-04-29.

B1 Instrumental LSF — KCWI header DISPSCAL = 0.294 (instrumental $\sigma$ in pixels) × CD3_3 = 1.000 Å/pix → FWHM_inst = 2.355 × 0.294 = 0.692 Å (constant in observed Å). At the rest-frame fit window (3879–4476 Å), $\sigma$_v_inst ≈ 12–14 km/s. Galaxy $\sigma \approx$ 270 km/s is resolved by ~20×. Diagnosed in `scripts/ifu_spectral_resolution.py`.

B2 $\sigma$_inst sensitivity — Sweep DISPSCAL × {0.5, 0.75, 1.0, 1.25, 1.5, 2.0}. Max $|\Delta\sigma|$ = 0.83 km/s across all SPS. Reason: ppxf’s `sps_lib` clips fwhm_diff$^2$ = (FWHM_gal$^2$ − FWHM_tem$^2$).clip(0) (sps_util.py:169). When templates are intrinsically broader (FSPS, EMILES at our band), no convolution is applied — $\sigma$ becomes insensitive to DISPSCAL. LSF is rock-solid. Script: `scripts/sigma_inst_sensitivity.py`

B3 SPS template frames — Located the Ca K minimum in each shipped template: - FSPS spectra_fsps_9.0.npz: Ca K at 3934.86 Å → +1.20 Å vs air rest = vacuum - EMILES spectra_emiles_9.0.npz: 3933.82 Å → +0.16 Å = air - XSL spectra_xsl_9.0.npz: V_sys closes at +9 km/s with air-conversion → air End-to-end V_sys closure (NOTES Test 3) confirms.

B4 V_sys air vs vacuum × 5 polynomial degrees — Run ppxf at degrees {15, 18, 21, 24, 27} for each SPS in both frames. $\Delta$V(air − vac) ≈ +83 km/s for FSPS, ≈ −82 km/s for EMILES & XSL, consistent within $\pm$2 km/s across degrees. Frame identification is robust.

B5 z × air-vac differential — Apply `vac_to_air` at observed wavelengths (KCWI 6500–7500 Å) → v_offset = −82.7 km/s. Apply at rest wavelengths (3879–4476 Å) → −84.5 km/s. Differential = 1.83 km/s — sub-budget. Either convention is defensible; we follow Cappellari's pattern (apply at obs).

B6 fwhm_gal_dict frame check — Confirmed `_prep_spectrum_for_ppxf` builds the dict as `{"lam": lam_gal_rest, "fwhm": fwhm_gal_rest}` in REST frame, matching `ppxf_example_high_redshift.py:99–101`. FWHM_obs = 0.692 Å → FWHM_rest = 0.413 Å at z=0.67564. `sps_util.py:167` interpolates this onto template `lam_temp` (also rest), then `varsmooth` pre-broadens.

C. SPS templates and frame-fix

C1 Frame-aware ppxf — `scripts/bootstrap_ppxf.py` exposes `frame_galaxy='auto'` which reads `SPS_NATIVE_FRAME = {fsps: vacuum, emiles: air, xsl: air}`. The galaxy spectrum is prepared in the matching frame (`util.vac_to_air()` applied as scalar median when needed, matching `ppxf_example_kinematics_sdss.py` :127-134). Diagnosed Apr 2026 — pre-fix the pipeline always converted to air, producing a −90 km/s FSPS V_sys offset.

C2 V_sys closure — Same galaxy spectrum, swap only the frame. Pre-fix: FSPS −90, EMILES +3, XSL +9. Post-fix (frame-aware): FSPS −7, EMILES +3, XSL +9. Split-track collapsed from ~110 → ~15 km/s. NOTES Test 3.

C3 $\sigma$ shift from frame fix — Side-effect on $\sigma$: $\leq$5 km/s per SPS, $\leq$2 km/s on the 3-SPS pool. Carried as $\pm$5 km/s in the error budget (ironically dominated by the per-SPS shifts that pool out, but conservative).

C4 3-SPS pooled posterior — `scripts/final_sigma_e.py:pool_sps` concatenates per-SPS bootstrap $\sigma$ samples after subtracting per-SPS V_sys medians (so V offsets don't inflate the V posterior; $\sigma$ posteriors are concatenated raw — no V_sys correction needed for $\sigma$). The headline $\sigma$_e(<R_e) is the 16/50/84 percentile of this combined-SPS pool.

D. Aperture and I(r) framework

D1 §6cum cumulative I-weighted ppxf — For each spaxel inside R<R_max, build I-weight from the IFU 6500–7500 Å white-light band, drop arc-mask-flagged spaxels, normalize and sum to a single 1-D spectrum, fit ppxf. This is the headline path — matches KH13 / Greene+20 / SAURON / ATLAS3D / MaNGA $\sigma$_e definition (Cappellari+2006 eq. 1). Cross-references in `~/.claude/.../reference_cumulative_vs_annular_sigma_e.md`. Code: `scripts/final_sigma_e.py:run_aperture_sps`.

D2 §6cum vs §7 (annular) — §7 discrete Gültekin annular sum at R<R_safe gives 254.99 km/s; §7b flat-$\sigma$ extrapolation gives 271 km/s; both within 1$\sigma$ of §6cum's 267.82 km/s. §6cum is the headline because (a) it matches the literature definition, (b) it preserves LOSVD line-shape information that §7's moment-pooling discards, (c) bright-center I-weighting auto-suppresses arc contamination.

D3 I(r) shape sweep (10 shapes) — Hold the spaxel selection fixed (F200-raw mask, R<R_e), vary only the I-weight map. Shapes: 1. IFU_band — 6500-7500 Å (headline) 2. IFU_wl — full IFU white-light 3. F140W_raw — HST F140W reprojected 4. F200LP_raw — HST F200LP reprojected 5. F140W_arcmasked — F140W with arc zeroed 6. F200LP_arcmasked — F200LP with own arc-mask zeroed 7. F140W_1Dcog — annular-mean F140W → interp1d to spaxels 8. F200LP_1Dcog — annular-mean F200LP → interp1d 9. F140W_Sersic2D — full 2D Sersic fit 10. F200LP_Sersic2D — full 2D Sersic fit (poor fit, n=0.30 — outlier)

Range excluding F200LP_Sersic2D outlier = 12.9 km/s → $\pm$13 km/s budget. Script: `scripts/run_isource_shape_sweep.py` Analysis: `scripts/analyze_isource_shape_sweep.py` Caches: `results/annular_bootstrap_07c_ishape/{shape}_{sps}.npz` Figure: `results/figures/nb07c_ishape_sweep.png`

D4 I-shape sweep refresh post-frame-fix — Original sweep at N=250 predated the frame fix. Re-ran at N=500 with `frame_galaxy='auto'` on 2026-04-29. Per-SPS $\sigma$ shifts <0.5 km/s; spread unchanged at 12.9 km/s; $\pm$13 budget validated. Pre-frame caches preserved at `results/annular_bootstrap_07c_ishape_oldframe/`. NOTES addendum (d).

D5 Aperture choice — Three apertures considered: R<R_e/8 (=0.288"), R<R_e/2 (=1.152"), R<R_e (=2.305"). R<R_e/8 dropped because it sits inside half the seeing FWHM (=0.64") — only 3 spaxels, and the inner $\sigma$ would underestimate due to seeing-broadened sampling. R<R_e is the paper headline; R<R_e/2 is quoted as a gradient diagnostic.

D6 Equal-N vs equal-width annular binning — For the §7 cross-check only. Equal-N inside R_safe = 3R_e/4 + 1 outer flagged bin (`notebooks/07c`) gives balanced bootstrap variance per bin and isolates arc contamination in the outer bin. Equal-width annuli (`notebooks/07b`, 5 bins) shifted $\sigma$_e by ~10 km/s — at the SPS-systematic level ($\pm$24). Equal-N is the §7 paper choice.

D7 R_e source systematic (mean vs F140W vs F200LP vs Ca H+K + G) — The headline integrates the I-weighted aperture spectrum inside the mean F140W+F200LP masked CoG R_e (=2.305"). Three alternative R_e definitions test sensitivity to that choice (Track A masked, N=500):

| R_e source | R_e ["] | $\sigma$_e [km/s] | $\Delta\sigma$_e |
|---|---|---|---|
| mean (paper) | 2.305 | 267.82 | — |
| F140W only | 2.168 | 264.87 | −2.96 |
| F200LP only | 2.441 | 275.86 | +8.04 |
| Ca H+K + G-band depth | 2.866 | 281.74 | +13.92 |

Spread = 16.9 km/s — at the mask-budget level ($\pm$16) but sub-budget vs the total $\pm$32. $\sigma$_e rises monotonically with R_e (more outer-bulge spaxels in the I-weighted aperture). The Ca H+K + G-band absorption- depth I-map (rest 3925-3942, 3960-3976, 4297-4313 Å, summed `(continuum − flux)` per spaxel — see `cahk_g_line_depth_map` in `scripts/final_sigma_e.py`) is the most stellar-only of the four; its larger R_e reflects that the deflector light extends further than the F200LP UV-leaning band suggests. Code: `scripts/final_sigma_e.py` §7 + `cahk_g_line_depth_map`. Display: `notebooks/09 §7b` + `results/figures/nb09_re_source_systematic.png`. Caches: `results/final_sigma_e_paper/Re_{F140W,F200LP,CaHK}_{sps}_N500.npz`.

E. Mask treatment

Masking vs no-masking — pro/con summary

| Track | Pro | Con |
|---|---|---|
| Hard mask (w=0.0, headline) | Removes arc contamination; matches literature convention; F200LP mask is purpose-built for this lens | Drops 38 of 184 spaxels inside R<R_e (~21% by count, ~27% by I-weight) $\rightarrow$ lower S/N; hard boundary may over-clip a few edge spaxels |
| No mask (w=1.0, sensitivity) | Maximum S/N; no boundary artifacts | Arc adds low-velocity continuum dilution $\rightarrow$ $\sigma$ biased low by 15-17 km/s ($\Delta$_mask = −16 km/s at R<R_e) |
| Soft mask (w=0.5) | Smooth middle ground | Super-linear in w (quadratic c=+7.3); threshold-dominated bias $\rightarrow$ first 25% of arc weight already captures 34% of total mask sensitivity. Arbitrary weight choice; not a literature convention |
| Mask dilation (E6) | — | Inflates $\sigma$ via noise (dropped spaxels are S/N contributors, not bias contributors). Negative finding — do not dilate |

The 16 km/s no-mask $\Delta$ is propagated into the error budget as `σ_mask = ±16`, so the choice is bounded, not assumed.

E1 F200LP arc mask reprojection — F200LP `_mask.fits` (HST 0.04"/pix, arc-tuned) is reprojected to the IFU 0.30"/spax grid via `scipy.ndimage.map_coordinates(order=0)` (nearest- neighbor). 69 of 10000 spaxels (0.69%) flagged. Inside R<R_e: 38 of 184 spaxels in the aperture, ~21% — but only ~27% of the I-weight (arc spaxels are mostly off-center).

E2 Hard-mask (w=0.0) headline — track A. E3 No-mask (w=1.0) sensitivity — track B. $\Delta$_mask = −15.0 km/s. E4 Soft-mask single point (w=0.5) — track C. $\Delta$_soft = −9.3 km/s. E5 5-point mask-weight sweep (w$\in${0,0.25,0.5,0.75,1}) — Per-step $\sigma$ drops at R<R_e: 5.0 $\rightarrow$ 4.3 $\rightarrow$ 3.5 $\rightarrow$ 2.2 km/s. Quadratic c = +7.3 (concave-up, super-linear). Threshold-dominated bias: a few heavily-arc-contaminated spaxels carry most of the bias. First 25% of arc weight captures 34% of the total mask sensitivity. Scripts: `scripts/soft_mask_track.py`, `scripts/analyze_mask_weight_sweep.py` Figure: `results/figures/nb09_mask_weight_sweep.png` NOTES addendums (b) and (c).

E6 Mask dilation — Tested whether expanding the F200 mask further reduces $\sigma$. Negative finding: dilation inflates $\sigma$ via noise — the dropped spaxels are S/N contributors, not bias contributors. Memory: `~/.claude/.../project_nb07d_overmasking_finding.md`. Notebook: `nb07d`.

E7 Arc-spectrum subtraction sibling — Build an arc-only spectrum from F200-masked spaxels, fit $\alpha \times$ arc + $\beta \times$ deflector model, subtract. Result matches §6cum within 0.1 km/s $\rightarrow$ *residual* arc dilution (the small fraction that leaks through the F200 mask) is sub-dominant at production statistics. The F200 mask already does the heavy lifting. Note: this test compares "masked + arc-subtract" against "masked alone" — it does NOT directly validate the no-mask vs masked $\Delta$ mechanism (that's E8). Notebook: `nb07e`.

E8 Arc-as-sky mechanism test (rigorous test of the no-mask $\Delta$) — Run §6cum at the *no-mask* R<R_e aperture with the bright outer-arc spectrum passed to ppxf as a `sky` template (free-amplitude additive component, NOT convolved with the deflector LOSVD — physically appropriate since the arc is at z=1.302, decoupled from the deflector kinematics). At N=500, FSPS+EMILES+XSL pooled:

| Track | $\sigma$_e [km/s] | $\Delta$ from $\sigma$_A |
|---|---|---|
| A: masked headline | 267.82 | — |
| B: no-mask | 252.82 | −15.00 |
| D: no-mask + arc-sky (E8) | 274.64 | +6.82 |

Recovery fraction ($\sigma$_D − $\sigma$_B)/($\sigma$_A − $\sigma$_B) = 145% — i.e. arc-sky overshoots the masked headline by ~7 km/s. Interpretation:

- Continuum dilution explains the FULL no-mask $\Delta$. There is no detectable kinematic-blend or template-mix mechanism contributing in the *opposite* direction.
- The 7 km/s overshoot is consistent with the known caveat that the outer-arc-mask spaxels (R > 3R_e/4) still receive seeing-blurred deflector light at the few-percent level, so subtracting $\alpha \times$ arc oversubtracts a small amount of *real* deflector flux too. This inflates $\sigma$ slightly.
- The masked headline (Track A, paper number) remains the cleaner measurement because it doesn't have this oversubtraction artefact.
- $\alpha$_arc $\approx$ 0.31 across all three SPS — internally consistent.

Code: `scripts/run_07f_arc_sky.py` (uses `setup_ppxf_inputs_from_spectrum` + ppxf `sky=` argument). Display: `notebooks/07f_arc_sky_subtract.ipynb`. Caches: `results/arc_sky_07f/{sps}_N500.npz` + `results/arc_sky_07f.npz`. Figures: `results/figures/nb07f_{arc_template,recovery,posteriors}.png`.

## F. Centering

F1 5-center sweep — Tested HST_mean (paper choice), F140W only, F200LP only, IFU peak, arc-suppressed IFU peak. $\sigma$_e spread = 3.7 km/s. Carried as $\pm$4 km/s in the error budget. Notes: `NOTES_centering_investigation_2026-04-27.md`.

F2 F140W vs F200LP centroid — $\Delta$ = 0.36" — driven by the F200LP arc dragging that filter's centroid toward the arc. F140W is the cleaner bulge centroid; HST-mean is the robust compromise.

## G. Bootstrap and polynomial degree

G1 Wild bootstrap — Hybrid Rademacher sign-flip $\times$ local-residual scaling (75-pix rolling window). For each iteration: residual = (galaxy − bestfit) $\times$ sign, scaled by local $\sigma$. Re-fit, record V/$\sigma$. Implemented in `scripts/bootstrap_ppxf.py:run_bootstrap_single_degree`.

G2 Parallel bootstrap — joblib + threadpoolctl pinning BLAS=1 per worker (otherwise BLAS thread oversubscription pathology — saw 4-hour runtimes). 8 workers gives ~6 min for 6 fits at N=500. Script: `scripts/bootstrap_ppxf_parallel.py:run_bootstrap_single_degree_parallel`.

G3 Polynomial degree sweep — Degrees 15-29 (15 values) per fit. Diagnostic: $\sigma$ vs degree should be flat within bootstrap envelope; monotonic trend indicates the polynomial is absorbing LOSVD signal. Plotted at `nb09 §6.5` and saved to `results/figures/nb09_sigma_vs_degree.png`. Saturation confirmed at this degree range — picked from earlier `notebooks/03c` analysis.

G4 N=50 vs N=500 — N=50 smoke and N=500 production agree on $\sigma$_e to within 1 km/s. N=50 is ~6 min, N=500 is ~50 min. Production headline is N=500; the smoke is for quick rebuilds.

## H. Cross-checks

H1 §7 discrete Gültekin — `notebooks/07c §7`. Annular sum $\sigma\_e^2(<R) = \Sigma F\_j (V\_j^2 + \sigma\_j^2) / \Sigma F\_j$ with arc-filter on outer annulus. Result: 254.99 −24.2/+28.4 km/s. <1$\sigma$ from §6cum.

H2 §7b flat-$\sigma$ extrapolation — `notebooks/07c §7b`. Push the §7 sum into the outer annulus assuming $\sigma$_outer = $\sigma$ at the last clean annulus. Result: 271 −33/+35 km/s. Bracket on the §7 result.

H3 F200 mask sensitivity (independent) — `results/annular_bootstrap_07c_nomask/`. Same §6cum pipeline with mask off. $\sigma$_e(<R_e) = 250.96 km/s. $\Delta$ = −16.4 km/s. Independently confirms the nb09 mask-weight sweep at w=1.0.

## I. Earlier sigma-discrepancy work (**`notebooks/05x`**)

The 21-test diagnostic notebook resolving the $\sigma \approx$ 301 km/s (190-spaxel box, NB01) vs $\sigma \approx$ 210 km/s (inner circular aperture, NB05) discrepancy. Resolution: $\sigma$ depends on aperture; the 190-spaxel box is contaminated by the lensed arc. This work motivated the cumulative-I path (Tests 7-13 explored masking variants, Tests 14-16 went radial, Tests 18-19 tried binning approaches). PowerBin (Test 19) was rejected. The headline analysis moved to circular apertures + cumulative I-weighting (nb06 $\rightarrow$ nb07c $\rightarrow$ nb09).

## 3. File index

### Documents (top-level)

| File | Purpose |
|---|---|
| `TESTS_AND_DIAGNOSTICS.md` | This file — master test index |
| `NOTES_nb09_frame_fix_and_final_sigma_e_2026-04-28.md` | Frame fix derivation + addenda (a)–(d) |
| `HANDOVER.md` | Apr 27 $\sigma$_e analysis snapshot (pre-frame-fix; superseded by nb09) |
| `HANDOFF_Ir_sweep_2026-04-28.md` | I(r) sweep design doc (consumed; sweep complete) |
| `NOTES_methodology_2026-04-27.md` | Cube provenance + earlier methodology |
| `NOTES_centering_investigation_2026-04-27.md` | 5-center sweep result + $\pm$4 km/s budget |
| `CLAUDE.md` | Repo conventions, key results summary |
| `README.md` | High-level project description |

### Notebooks (paper-ready / current)

| Notebook | Purpose |
|---|---|
| `notebooks/09_final_sigma_e_paper.ipynb` | Paper-ready $\sigma$_e (headline 268 $\pm$ 32 km/s) |
| `notebooks/04_redshift_verification.ipynb` | z = 0.67564 line-fit |
| `notebooks/02_streamlined_Bagpipes_SED.ipynb` | log M$\star$ = 11.33 |
| `notebooks/08_sersic_total_photometry.ipynb` | Sersic-total photometry cross-check |
| `notebooks/07c_sigma_e_equalN.ipynb` | §6cum + §7 equal-N annular cross-check |
| `notebooks/07e_sigma_e_arc_subtract.ipynb` | Arc-subtraction sibling cross-check |
| `notebooks/05x_sigma_discrepancy_diagnostic.ipynb` | Tests 1-21 (foundational diagnostics) |

### Notebooks (history / reference, not paper-driving)

`01_*`, `02_*`, `03_bootstrap_ppxf_errors`, `03b_*`, `03c_*`, `05_*`, `06_*`, `07_*`, `07a_*`, `07b_*`, `07d_*` — kept for the history of the analysis.

### Scripts (production)

| Script | Purpose |
|---|---|
| `scripts/final_sigma_e.py` | nb09 production driver — runs all 3 mask tracks at N=500 |
| `scripts/build_nb09.py` | Generates the paper notebook from the saved artifacts |
| `scripts/bootstrap_ppxf.py` | Frame-aware ppxf input prep + wild bootstrap |
| `scripts/bootstrap_ppxf_parallel.py` | joblib parallel bootstrap (BLAS=1 per worker) |
| `scripts/measure_Re.py` | R_e via masked CoG (multi-band, multi-strategy) |
| `scripts/redshift_verify.py` | Line-fitting redshift module |
| `scripts/ifu_spectral_resolution.py` | LSF audit |

### Scripts (sweeps and audits)

| Script | Purpose |
|---|---|
| `scripts/audit_ppxf_methodology.py` | 4 orthogonal correctness audits |
| `scripts/sigma_inst_sensitivity.py` | DISPSCAL $\times$ {0.5..2.0} sweep |
| `scripts/run_isource_shape_sweep.py` | 10-shape I(r) sweep |
| `scripts/analyze_isource_shape_sweep.py` | I-shape sweep analysis + figure |
| `scripts/soft_mask_track.py` | Mask-weight runs (CLI `--weight`) |
| `scripts/analyze_mask_weight_sweep.py` | 5-point sweep analysis + linearity fit |
| `scripts/regen_s6cum_nomask_diagnostic.py` | Centering investigation |
| `scripts/relock_nomask_Re_N500.py` | One-off N=500 relock for the no-mask track |

Result files (key)

| File | Contents |
|---|---|
| `results/final_sigma_e_paper.npz` | nb09 headline + per-SPS + 3-track summaries |
| `results/final_sigma_e_paper/` | Per-(aperture, SPS, mask_weight) caches |
| `results/sigma_e_ishape_sweep.npz` | I-shape sweep summary |
| `results/sigma_e_ishape_sweep_oldframe.npz` | Pre-frame-fix I-shape (audit) |
| `results/annular_bootstrap_07c_ishape/` | Per-(shape, SPS) I-shape caches (N=500) |
| `results/annular_bootstrap_07c_ishape_oldframe/` | Pre-frame-fix I-shape caches (N=250) |
| `results/annular_bootstrap_07c_nomask/` | §6cum nomask caches |
| `results/mask_weight_sweep.npz` | 5-point mask-weight sweep summary |
| `results/ppxf_methodology_audit.npz` | 4-audit results |
| `results/figures/nb09_*.png` | nb09 paper figures |
| `results/figures/nb07c_ishape_sweep.png` | I-shape sweep figure |

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## 4. Reproduction recipes

```
conda activate ISMgas

# === Headline pipeline (~50 min at N=500) ===
python scripts/final_sigma_e.py --n_bootstrap 500
python scripts/build_nb09.py
jupyter nbconvert --to notebook --execute --inplace notebooks/09_final_sigma_e_paper.ipynb

# === Methodology audits (~3 min) ===
python scripts/audit_ppxf_methodology.py
python scripts/sigma_inst_sensitivity.py

# === I-shape sweep (~50 min at N=500) ===
python scripts/run_isource_shape_sweep.py --n-bootstrap 500 --n-jobs 8
python scripts/analyze_isource_shape_sweep.py

# === Mask-weight sweep (~25 min total) ===
# (w=0 and w=1 are produced by final_sigma_e.py; only need 0.25/0.5/0.75)
python scripts/soft_mask_track.py --weight 0.25
python scripts/soft_mask_track.py --weight 0.50
python scripts/soft_mask_track.py --weight 0.75
python scripts/analyze_mask_weight_sweep.py

# === Smoke run (~6 min, useful for rebuilds) ===
python scripts/final_sigma_e.py --n_bootstrap 50
```

Caches are keyed by N_bootstrap and mask_weight in the filename — re-runs at the same N skip cached fits unless `--force` is passed.

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## 5. What's intentionally NOT done

- Re-run nb09 at higher N (e.g. N=2000): the pooled posterior is already smooth; SPS systematics dominate the residual statistical variance.
- Re-fit Sersic2D F200LP: the n=0.30 fit is non-physical due to UV-arc domination; the F140W Sersic fit is the trustworthy one. F200LP_Sersic2D is the I-shape outlier *by design* and is excluded from the budget.
- Voronoi binning — Test 18 / nb05x — abandoned; not pursued at N=500.
- PowerBin spatial binning — Test 19 / nb05x — rejected (outer-bin $\sigma > 800$ km/s); not pursued.
- Per-spaxel rotation map at production statistics — no rotation detected at the seeing limit; not paper-driving.
- Full nb07a Sersic-I revisit at the post-frame-fix headline — superseded by the I-shape sweep (D3/D4) which covers the Sersic2D shape.

*See `~/.claude/projects/.../memory/MEMORY.md` for compressed cross-session references, including: - `reference_cumulative_vs_annular_sigma_e.md` — §6cum vs §7 design choice - `reference_sps_systematic.md` — per-SPS V_sys + frame-fix details - `reference_ppxf_vacair_handling.md` — vac/air conversion methodology - `reference_kcwi_data_properties.md` — KCWI cube + LSF + paper boilerplate - `project_nb09_final_sigma_e.md` — current headline status*